

A More Details on CARNet

The overall architecture of CARNet is described in Algorithm 1, while the proposed Multihead core aggregation mechanism is detailed in Algorithm 2. The pseudocode formalizes the forward computation of CARNet, highlighting how cycle information is integrated with variate embeddings through Cycle-Conditioned Core Aggregation and redistribution across multiple layers. This formulation clarifies the interaction between global periodic representations and cross-variate dependencies, and illustrates how the proposed modules are seamlessly embedded within the SOFTS-style star-shaped framework.

Algorithm 1 Pseudocode of CARNet

Require: Variate embedding $\mathbf{X}' \in \mathbb{R}^{C \times d}$, phase-aligned cycle $\mathbf{Z} \in \mathbb{R}^{C \times T}$

- 1: **for** $\ell = 1$ **to** L **do**
- 2: $\mathbf{Z}' \leftarrow \mathcal{L}_Z(\mathbf{Z})$ // Cycle projection, $\mathbf{Z}' \in \mathbb{R}^{C \times d}$
- 3: $\mathbf{V} \leftarrow \mathcal{F}(\mathbf{X}', \mathbf{Z}')$ // Feature fusion (concat), $\mathbf{V} \in \mathbb{R}^{C \times 2d}$
- 4: $\mathbf{V}' \leftarrow \psi(\mathcal{L}_V(\psi(\mathbf{V})))$ // Cycle-conditioned variate embedding, $\mathbf{V}' \in \mathbb{R}^{C \times d}$
- 5: $\boldsymbol{\theta} \leftarrow \text{MHCA}(\mathbf{V}')$ // Core aggregation, $\boldsymbol{\theta} \in \mathbb{R}^{C \times d'}$
- 6: $\boldsymbol{\theta}' \leftarrow \beta(\mathcal{P}(\boldsymbol{\theta}))$ // Pool + broadcast, $\boldsymbol{\theta}' \in \mathbb{R}^{C \times d'}$
- 7: $\tilde{\mathbf{V}} \leftarrow \mathcal{F}(\mathbf{X}', \mathbf{Z}', \boldsymbol{\theta}')$ // Redistribution fusion, $\tilde{\mathbf{V}} \in \mathbb{R}^{C \times (2d+d')}$
- 8: $\mathbf{G} \leftarrow \mathcal{L}_F(\psi(\mathcal{L}_{\tilde{V}}(\tilde{\mathbf{V}})))$ // Redistributed features, $\mathbf{G} \in \mathbb{R}^{C \times d}$
- 9: $\mathbf{O} \leftarrow \text{FFN}(\mathbf{X}' + \mathbf{G})$ // Output representation, $\mathbf{O} \in \mathbb{R}^{C \times d}$
- 10: $\mathbf{X}' \leftarrow \mathbf{O}$ // Update for next layer
- 11: **end for**
- 12: **return** $\mathbf{X}'$

Algorithm 2 Pseudocode of Multihead Core Aggregation (MHCA)

Require: Input $\mathbf{V}' \in \mathbb{R}^{C \times d}$, number of heads h

- 1: $\mathbf{H} \leftarrow \alpha(\mathcal{L}_H(\mathbf{V}'))$ // Head-wise decomposition, $\mathbf{H} \in \mathbb{R}^{C \times h \times d_{\text{head}}}$
- 2: **for** $i = 1$ **to** h **do**
- 3: $\boldsymbol{\theta}_i \leftarrow \psi(\omega(\mathbf{H}_i))$ // Head-wise core extraction, $\boldsymbol{\theta}_i \in \mathbb{R}^{C \times d'_{\text{head}}}$
- 4: **end for**
- 5: $\tilde{\boldsymbol{\theta}} \leftarrow \alpha([\boldsymbol{\theta}_1; \dots; \boldsymbol{\theta}_h])$ // Merge heads, $\tilde{\boldsymbol{\theta}} \in \mathbb{R}^{C \times d'}$
- 6: $\boldsymbol{\theta} \leftarrow \mathcal{L}_{\theta}(\tilde{\boldsymbol{\theta}})$ // Unified Multihead core, $\boldsymbol{\theta} \in \mathbb{R}^{C \times d'}$
- 7: **return** $\boldsymbol{\theta}$

B Dataset Description

We evaluate our method on twelve widely used real-world multivariate time-series datasets spanning multiple application domains, including electricity con-

sumption, energy generation, traffic flow, and weather forecasting. The benchmarks include the ETT datasets, PEMS traffic datasets, and the Electricity, Solar, Traffic, and Weather datasets.

Table 1. Detailed dataset description with standardized train/validation/test splits following prior benchmark settings [1,3,2,4].

Dataset	Domain	Interval	Channels	Timesteps	Dataset Split
ETTh1	Electricity	1h	7	14400	(8545, 2881, 2881)
ETTh2	Electricity	1h	7	14400	(8545, 2881, 2881)
ETTM1	Electricity	15m	7	57600	(34465, 11521, 11521)
ETTM2	Electricity	15m	7	57600	(34465, 11521, 11521)
ECL	Electricity	1h	321	26304	(18317, 2633, 5261)
Solar	Energy	10m	137	52560	(36601, 5161, 10417)
Traffic	Transportation	1h	862	17544	(12185, 1757, 3509)
Weather	Weather	10m	21	52696	(36792, 5271, 10540)
PEMS03	Transportation	5m	358	26208	(15617, 5135, 5135)
PEMS04	Transportation	5m	307	16992	(10172, 3375, 3375)
PEMS07	Transportation	5m	883	28224	(16911, 5622, 5622)
PEMS08	Transportation	5m	170	17856	(10690, 3548, 3548)

¹ <https://github.com/zhouhaoyi/ETDataset>

² <https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014>

³ <https://pems.dot.ca.gov>

Following standard experimental settings, the forecasting horizons for the PEMS datasets are selected from {12, 24, 48, 96}, while horizons of {96, 192, 336, 720} are used for the remaining datasets. As summarized in Table 1, the datasets also differ substantially in temporal resolution, dimensionality, total timesteps, and standardized training/validation/test splits, providing a comprehensive benchmark for long-term forecasting.

The **ETT** dataset¹ consists of four electricity transformer temperature subsets, including two hourly datasets (ETTh1, ETTh2) and two 15-minute datasets (ETTM1, ETTM2), each with seven variables. **Traffic** records hourly road occupancy rates collected from sensors on San Francisco freeways. **Electricity (ECL)**² provides hourly electricity consumption data from 321 clients. **Weather** contains 21 meteorological variables recorded every 10 minutes in Germany. **Solar-Energy** includes 10-minute solar power generation data from 137 photovoltaic plants. **PEMS**³ comprises 5-minute traffic flow measurements collected from California traffic networks.

C Experiment Details

All experiments are conducted on a single NVIDIA GeForce RTX 5060 GPU with 8 GB VRAM. Performance comparisons among different methods are evaluated using two primary metrics: Mean Squared Error (MSE) and Mean Absolute Error (MAE). The training, validation, and test splits are kept consistent with TQNet [1]. Specifically, the data are divided using a 6:2:2 ratio for the ETT and PEMS series datasets and a 7:1:2 ratio for the remaining datasets. CARNet is trained for 30 epochs on all datasets except ETTm1 and ETTm2, where the number of epochs is limited to 7, with a patience counter of 3 applied across all datasets. A fixed random seed of 2024 is used to ensure reproducibility.

D Additional Prediction Visualizations

To further illustrate the forecasting behavior of CARNet, we provide additional qualitative prediction visualizations on two datasets, **PEMS08** and **ETTM1**. For both datasets, we consider an input sequence length of 96 and a prediction horizon of 96, and compare CARNet against representative strong baselines, including TimeXer and TQNet. Figures 1 and 2 present the corresponding prediction results on PEMS08 and ETTm1, respectively.

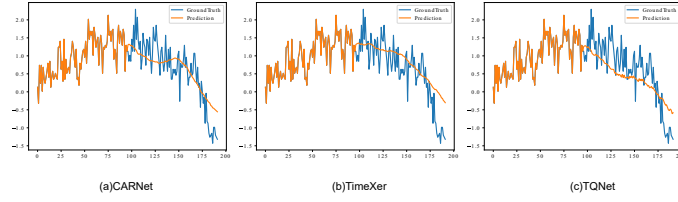


Fig. 1. Visualization of forecasting results on the PEMS08 dataset with look-back window 96 and prediction horizon 96.

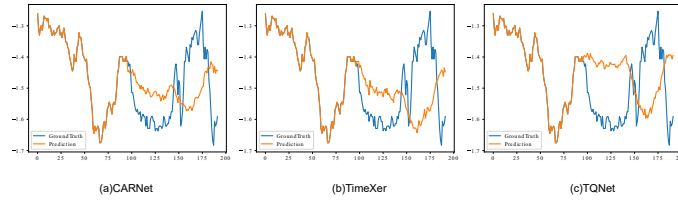


Fig. 2. Visualization of forecasting results on the ETTm1 dataset with look-back window 96 and prediction horizon 96.

E Full Results of Component Ablation

To evaluate the contribution of individual components, we conduct ablation studies by removing each module from the full model and measuring the resulting performance degradation. Specifically, we remove (i) the Cycle-Conditioned Core Aggregation, which excludes the cycle projection and its fusion with the variate embeddings, (ii) the Multihead Core Aggregation module, and (iii) the Cycle-Conditioned Core Redistribution, which redistributes the cycle projection together with the cycle-conditioned core to the variate embeddings. Table 2 reports the results across six datasets and four prediction horizons. The results indicate that each component contributes meaningfully to the overall forecasting performance.

Table 2. Full results of the component ablation study.

Dataset	Pred	cSOFTS		w/o MHCA		w/o CCCA		w/o CCCR	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	0.304	0.348	0.308	0.352	0.311	0.355	0.316	0.355
	192	0.354	0.379	0.357	0.381	0.359	0.382	0.361	0.380
	336	0.388	0.401	0.394	0.404	0.390	0.404	0.398	0.404
	720	0.454	0.438	0.460	0.443	0.457	0.441	0.459	0.439
	Avg	0.375	0.392	0.380	0.395	0.379	0.396	0.384	0.395
Solar	96	0.177	0.231	0.178	0.245	0.176	0.237	0.189	0.238
	192	0.198	0.250	0.206	0.263	0.207	0.267	0.202	0.254
	336	0.208	0.260	0.213	0.268	0.222	0.274	0.236	0.282
	720	0.209	0.256	0.214	0.264	0.230	0.280	0.215	0.261
	Avg	0.198	0.249	0.203	0.260	0.209	0.265	0.211	0.259
Weather	96	0.155	0.200	0.157	0.203	0.157	0.202	0.169	0.206
	192	0.203	0.244	0.207	0.247	0.205	0.245	0.216	0.249
	336	0.265	0.289	0.264	0.288	0.263	0.290	0.276	0.294
	720	0.344	0.342	0.344	0.342	0.346	0.344	0.351	0.344
	Avg	0.242	0.269	0.243	0.270	0.243	0.270	0.253	0.273
PEMS04	12	0.064	0.161	0.067	0.166	0.067	0.163	0.074	0.175
	24	0.071	0.170	0.077	0.167	0.075	0.172	0.089	0.194
	48	0.083	0.185	0.086	0.187	0.085	0.186	0.112	0.220
	96	0.099	0.203	0.102	0.206	0.100	0.205	0.138	0.247
	Avg	0.079	0.180	0.083	0.182	0.082	0.182	0.103	0.209
PEMS08	12	0.069	0.164	0.069	0.165	0.069	0.165	0.076	0.173
	24	0.087	0.185	0.090	0.187	0.092	0.188	0.105	0.202
	48	0.125	0.212	0.128	0.213	0.132	0.222	0.167	0.253
	96	0.204	0.250	0.234	0.262	0.206	0.248	0.265	0.320
	Avg	0.121	0.203	0.130	0.207	0.125	0.206	0.153	0.237
ECL	96	0.131	0.226	0.132	0.226	0.131	0.226	0.147	0.237
	192	0.149	0.244	0.152	0.247	0.151	0.244	0.160	0.251
	336	0.164	0.260	0.168	0.264	0.165	0.262	0.178	0.270
	720	0.188	0.285	0.194	0.288	0.192	0.286	0.208	0.297
	Avg	0.158	0.254	0.162	0.256	0.160	0.255	0.173	0.264

F Results Under Different Random Seeds

To further examine the robustness of CARNet, we conduct experiments on five representative datasets using multiple random initializations. For each dataset, results are reported across three random seeds for four forecasting horizons, and we summarize performance using the mean and standard deviation. As presented

in Table 3, CARNet consistently achieves low standard deviation across a wide range of settings, indicating stable training dynamics and reliable forecasting performance. Overall, these results demonstrate that the effectiveness of CARNet is not dependent on a particular random seed and generalizes consistently across datasets with diverse temporal characteristics.

Table 3. Performance of CARNet under different random seeds. Mean represents the average value, and Std denotes the standard deviation.

Dataset	Pred	Random Seed									
		2024		2025		2026		Mean		Std	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Electricity	96	0.131	0.226	0.132	0.228	0.131	0.227	0.1313	0.2270	0.0005	0.0008
	192	0.151	0.244	0.150	0.245	0.150	0.245	0.1503	0.2447	0.0005	0.0005
	336	0.166	0.260	0.166	0.265	0.165	0.264	0.1657	0.2630	0.0005	0.0022
	720	0.190	0.285	0.192	0.289	0.198	0.293	0.1933	0.2890	0.0034	0.0033
Weather	96	0.155	0.200	0.155	0.200	0.155	0.200	0.1550	0.2000	0.0000	0.0000
	192	0.203	0.244	0.203	0.245	0.205	0.245	0.2037	0.2447	0.0009	0.0005
	336	0.265	0.289	0.267	0.292	0.265	0.289	0.2657	0.2900	0.0009	0.0014
	720	0.344	0.342	0.350	0.344	0.345	0.343	0.3463	0.3430	0.0026	0.0008
Solar	96	0.177	0.231	0.185	0.243	0.190	0.241	0.1840	0.2383	0.0053	0.0053
	192	0.198	0.250	0.200	0.251	0.210	0.265	0.2027	0.2553	0.0052	0.0071
	336	0.208	0.260	0.213	0.261	0.204	0.257	0.2083	0.2593	0.0037	0.0017
	720	0.209	0.256	0.212	0.264	0.236	0.282	0.2190	0.2673	0.0121	0.0109
PEMS04	12	0.064	0.161	0.064	0.161	0.063	0.161	0.0637	0.1610	0.0005	0.0000
	24	0.071	0.170	0.070	0.169	0.072	0.170	0.0710	0.1697	0.0008	0.0005
	48	0.083	0.185	0.084	0.187	0.085	0.188	0.0840	0.1867	0.0008	0.0012
	96	0.099	0.203	0.097	0.202	0.097	0.202	0.0977	0.2023	0.0009	0.0005
PEMS08	12	0.069	0.164	0.069	0.166	0.069	0.164	0.0690	0.1647	0.0000	0.0009
	24	0.087	0.185	0.092	0.188	0.092	0.191	0.0903	0.1880	0.0024	0.0024
	48	0.125	0.212	0.129	0.214	0.129	0.214	0.1277	0.2133	0.0019	0.0009
	96	0.204	0.250	0.225	0.273	0.208	0.252	0.2123	0.2583	0.0092	0.0104

References

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